File contents

- Curated MSA: GH19_Curated_MSA.fasta
- Raw GH19 seqs from plant: GH19_plant.txt
- After CD-hit 50% cutoff: GH19_labbeled.txt
- Library Seqs: GH19_Ancestors_Library.fasta
- Antifungal GH19 sequences: Antifungal_Chitinases.fasta
- Slides used for the meeting: ProtMamba_InDels.pptx

Loop regions definition

Loop regions are difined by structural and sequence comparison between loopless from Gemmabryum coronatum (Uniprot ID: A9ZSX9; residues 25-228) and loopful GH19 sequences from Secale cereale_ (Uniprot: Q9FRV0; residues 24-266). Curated MSA is initially aligned by MAFFT, manually curated based on profile alignment of the structural alignment of loopless and loopful GH19 chitinase.

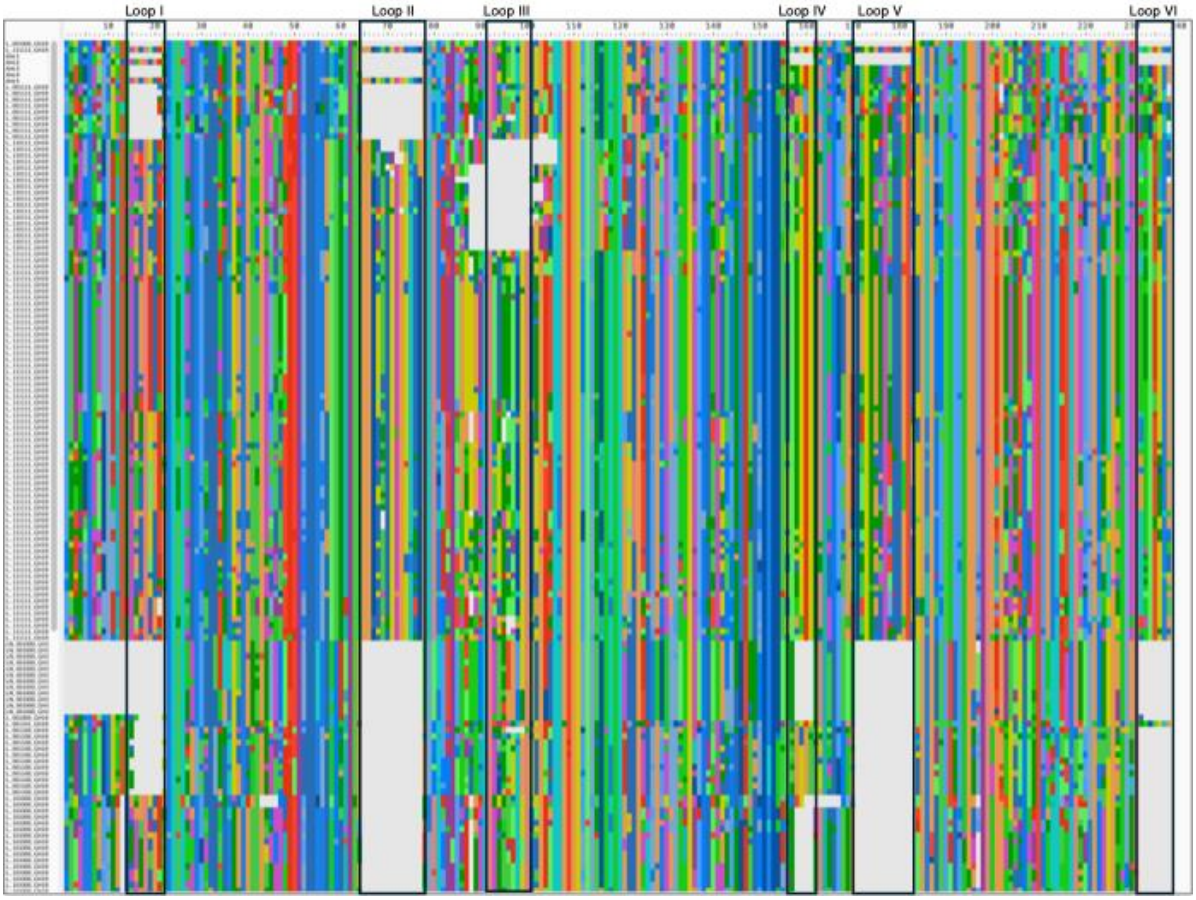


Fig. 1. Screenshot of curated MSA

In curated MSA, the first seq is loopless GH19, the 2nd seq is loopful GH19.

Loop I (pos 15 to 21, seven amino acid length)

Loop II (pos 65 to 77, 13 aa length)

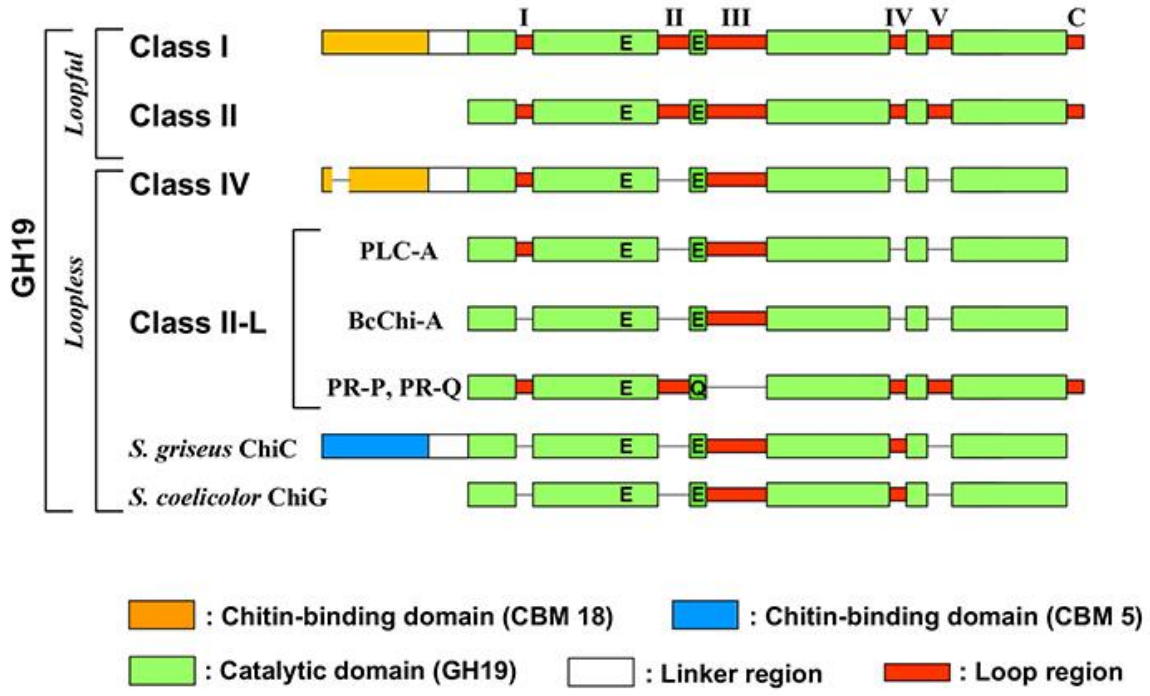
Loop III (pos 92 to 100, eight aa length)

Loop IV (pos 157 to 161, five aa length)

Loop V (pos 171 to 182, 12 aa length)

Loop VI (pos 232 to 238, seven aa length)

Background info on GH19 family classification



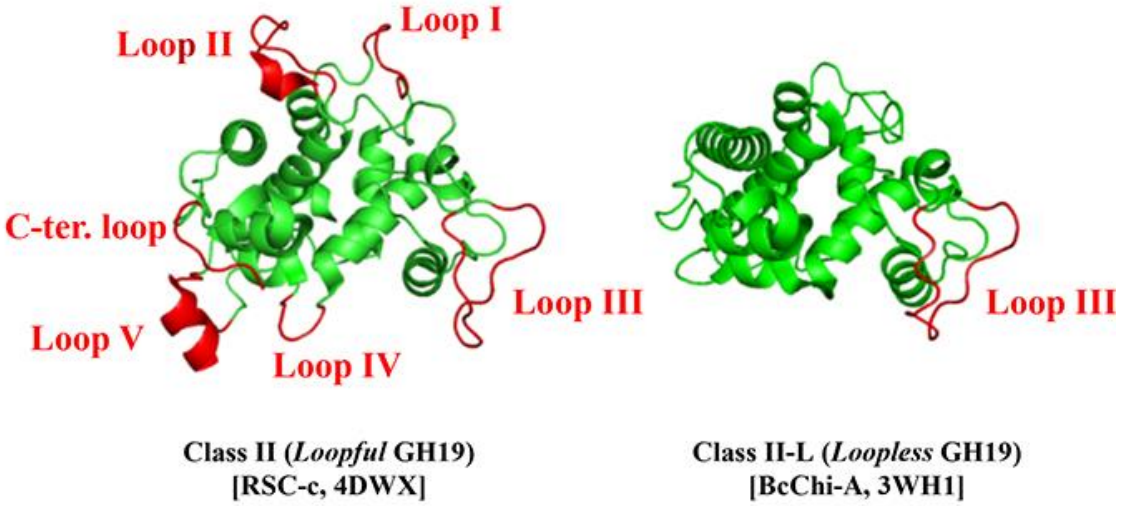


Fig. 2. Structure of GH19 chitinase from the ref (DOI:10.32285/glycoforum.26A4)

GH19 chitinases are divided into class I, II, and IV. Class I chitinases consist of an N-terminal chitin-binding domain, belonging to carbohydrate-binding module family 18 (CBM18), and a GH19 catalytic domain. Class II chitinases have only a catalytic domain homologous to that of class I chitinases. Class IV chitinases share sequence similarity with class I chitinases, but show the defference in a molecular mass due to the deletion of loop regions.**Note that we are focusing on only GH19 catalytic domain sequences. So, in sequence collections (raw data), some seq have CBM domain, but the Curated MSA containing only GH19 catalytic domain.**

What we want to know

- Can ProtMamba generate loop II regions and predict functioanl seq, and are these seqs are match with our experimental data?